

NEF ABUNDANCE: THREEFOLD BEHAVIOR

Theorem. *Let X be a smooth projective threefold over $\mathbb{C}$ such that $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor on X such that $n(X, D) = 3$ and $D^3 = 0$. Then*

$$H^0(X, \mathcal{O}_X(mD)) \neq 0 \quad \text{for some integer } m > 0.$$

The argument below is a source-based proof draft following [3]. The moving-kernel, slice, and correspondence arguments needed here are written out, following that manuscript and its companion [1]. Published inputs are cited with their hypotheses. This report remains a proof draft whose correctness has not been independently certified. We stop at nonvanishing.

1. PROOF FRAMEWORK

Setup. Fix an ample Cartier divisor A , and a positive rational number ϵ such that

$$(14\epsilon)^3 < \frac{1}{4}, \quad 80\epsilon < \frac{3}{4}.$$

Let t run through a fixed sequence of positive rational numbers tending to zero. Set $a_t = ((D + tA)^3)^{1/3}$, choose r_t to be the nearest positive integer to ϵ/a_t , and put

$$(1) \quad P_t = r_t(D + tA), \quad \rho_t = (P_t^3)^{1/3}.$$

Since $D^3 = 0$, these are ample rational Cartier divisors with

$$r_t \longrightarrow \infty, \quad \rho_t \longrightarrow \epsilon, \quad P_t^3 \longrightarrow \epsilon^3;$$

see Lemma 3.1. Every integer k used with $|kP_t|$ is positive and chosen so that kP_t is Cartier.

Let D_i and $P_{t,i}$ denote pullbacks from the two factors of $X \times X$, and then to the projective bundle

$$Z = \mathbb{P}_{X \times X}(\mathcal{O}_{X \times X}(D_1) \oplus \mathcal{O}_{X \times X}(D_2)), \quad \xi = c_1(\mathcal{O}_Z(1)).$$

For a positive integer Q , put $R_t = P_{t,1} + P_{t,2} + Q\xi$. The torus open is the complement of the two summand sections.

Suppose for contradiction that

$$(2) \quad \kappa(D) = -\infty, \quad \text{equivalently } H^0(X, \mathcal{O}_X(mD)) = 0 \text{ for every integer } m > 0.$$

Lemma 1.2 supplies a fixed Q and a large jet system for all sufficiently small t . Choose one such t for which Lemma 1.1 also holds, $r_t > Q + 1$, and $P_t^3 < (2\epsilon)^3$, and put $P = P_t$. At a very general point x , Lemma 1.1 gives the multiplicity bound for $E \in |kP|$ and an actual movable witness on the point blowup. Lemma 2.2 supplies a smooth complete intersection flag with final curve C satisfying $\mu_{\min}(\Omega_X^1|_C) \geq 0$. Fix the jet system, witness, and flag over $\mathbb{C}$.

After reduction to large characteristics p , Lemma 1.3 uses the jets to obtain generic rank p^6 and a nonzero member E_p of the determinant linear system. The diagonal rank is less than $p^6/4$; together with the movable witness, this gives

$$\frac{3}{4}p^6 < \text{mult}_{(x', x')}(E_p) \leq (80\epsilon + O(p^{-1}))p^6.$$

This contradicts $80\epsilon < 3/4$ for large p , proving the required nonvanishing.

The three lemmas below use the hypotheses of the theorem, the temporary assumption (2), and the auxiliary choices just made. Their proofs occupy Sections 3, 4, and 5, respectively. All sufficiently small t refer to the fixed sequence in (1). Write $\nu(D) = \max\{j : D^j A^{3-j} > 0\}$; here $\nu(D) \in \{1, 2\}$.

Lemma 1.1 (Scalar bound and movable witness). *For all sufficiently small t , at a very general point $x \in X$, every member $E \in |kP_t|$ satisfies*

$$\text{mult}_x(E) \leq 4k\rho_t.$$

For every fixed positive integer Q and all sufficiently small t , the point blowup $\pi_x : \hat{X} \rightarrow X$, with exceptional divisor J , admits an actual strongly movable class $\gamma = f_(A_1 A_2)$, on a smooth birational model with A_1, A_2 ample Cartier, such that*

$$J\gamma > 0, \quad \pi_x^*(P_t + \lambda D)\gamma \leq 40\epsilon J\gamma \quad (0 \leq \lambda \leq Q).$$

Lemma 1.2 (Sevenfold jets). *There is a fixed positive integer Q such that, for all sufficiently small t , there is a point z in the torus open of Z with $\varepsilon(R_t; z) > 8$. For some sufficiently divisible integer $k_0 > 0$ with $k_0 P_t$ Cartier, finitely many sections of $k_0 R_t$ generate jets through order $8k_0$ at z .*

Lemma 1.3 (Determinant comparison). *Fix Q as in Lemma 1.2, choose t with $r_t > Q + 1$ and $P_t^3 < (2\epsilon)^3$, and put $P = P_t$. Suppose a jet system as in Lemma 1.2, a movable witness as in Lemma 1.1, and a smooth complete intersection flag of successive classes $l_1 P, l_2 P$ with final curve C satisfying $\mu_{\min}(\Omega_X^1|_C) \geq 0$ are all fixed over $\mathbb{C}$. On smooth reductions in arbitrarily large characteristics p , the resulting evaluation map has generic rank p^6 and rank less than $p^6/4$ at (x', x') . Its determinant defines a nonzero member E_p of the determinant linear system on $Y' \times Y'$, where Y' is the relative Frobenius twist, satisfying*

$$\frac{3}{4}p^6 < \text{mult}_{(x', x')}(E_p) \leq (80\epsilon + O(p^{-1}))p^6.$$

Hence these fixed data cannot coexist.

2. COMMON PRELIMINARY INPUTS

The following auxiliary result is used inside the moving-centre and jet arguments. Its varieties have dimension at most two; it does not assume threefold nef nonvanishing.

Lemma 2.1. *Let (Y, B) be an auxiliary projective klt pair of dimension $f \in \{1, 2\}$, with $T = K_Y + B$ pseudo-effective, and let M be nef and rational Cartier. Then $T + M$ has an effective rational numerical representative. Moreover, if for some $a > 0$*

$$(T + aM) \cdot C > 0$$

for every integral curve through a sufficiently very general point, then $T + cM$ is big for every rational $c > 0$.

Proof. Numerical nonvanishing on surfaces is [5, Corollary C]; on curves it follows from the degree criterion. For the second assertion, the curve case is immediate. In dimension two run the ordinary $(K_Y + B)$ -minimal model program, after a small $\mathbb{Q}$ -factorialization. Its output is a morphism $g : Y \rightarrow Y_0$ with $U = K_{Y_0} + B_0$ nef. The pushforward $N = g_* M$ is nef. Negativity gives

$$T = g^* U + E_T, \quad M = g^* N - E_M,$$

where $E_T, E_M \geq 0$ are exceptional; E_T has positive coefficient on every contracted divisor. Thus, for small rational $s > 0$, $E_T - sE_M \geq 0$.

By [5, Corollary C], $U + sN$ is numerically equivalent to a semiample rational divisor. If it were not big, the resulting morphism would have positive-dimensional fibres. A general

fibre, or a general curve in such a fibre when the image is a point, avoids the finitely many points $g(\text{Exc } g)$. Since U and N are nef, this curve has both U - and N -degree zero. Its inverse image contradicts the stated full positivity of $T + aM$. Consequently $U + sN$, and hence $T + sM = g^*(U + sN) + E_T - sE_M$, is big. For $c > 0$, choose $0 < \lambda < \min\{1, c/s\}$ and write

$$T + cM = \lambda(T + sM) + (1 - \lambda)T + (c - \lambda s)M.$$

The last two summands are pseudo-effective, so the sum is big. This proves exactly the dimension-one and dimension-two inputs used in [3, Sections 3–5], without assuming threefold nef nonvanishing. $\square$

Lemma 2.2. *For the fixed ample rational divisor P , there is a smooth complete intersection flag with successive classes l_1P, l_2P and final curve C such that $\mu_{\min}(\Omega_X^1|_C) \geq 0$.*

Proof. Since $K_X \sim 0$, generic cotangent semipositivity [15, Theorem 2.1] makes Ω_X^1 semistable of slope zero for this polarization. Mehta–Ramanathan restriction [16, Theorem 6.1] gives the assertion for sufficiently large divisible l_1, l_2 . Bertini makes the flag smooth and integral. These results apply in arbitrary dimension; here only two restrictions are needed. $\square$

3. PROOF OF THE SCALAR AND MOVABLE-CURVE LEMMA

We prove Lemma 1.1. We first establish the normalization and moving-centre estimates, then obtain the scalar bound and its movable witness. The moving-centre tools will also be used in the proof of Lemma 1.2.

We retain the normalization, full jet kernels, parameter differentiation, and intersection argument of the general proof. All statements below use the full-nef, non-big setting of Section 1. Their arguments apply in arbitrary dimension; the output here is either a curve or a surface.

Lemma 3.1. *For every $\epsilon > 0$ and every sequence of positive rational numbers $t \rightarrow 0$, there are positive integers $r_t \rightarrow \infty$ such that*

$$P_t = r_t(D + tA), \quad P_t^3 \rightarrow \epsilon^3.$$

The divisor P_t is ample and rational Cartier; its Cartier multiples are chosen after this normalization.

Proof. Put $a_t = ((D + tA)^3)^{1/3}$. Then $a_t > 0$ and $a_t \rightarrow 0$. Choose r_t to be the nearest positive integer to ϵ/a_t . For sufficiently small t , $|r_t a_t - \epsilon| \leq a_t/2$, which proves the assertions. $\square$

Lemma 3.2. *Let W be smooth projective of dimension d , let H be a nef Cartier divisor on W , and let $\Lambda \subseteq |H|$ be a nonempty linear system with base ideal $\mathfrak{b}$. Let V be an integral component of its base locus, of codimension $c \geq 1$. If*

$$\mathfrak{b}\mathcal{O}_{W, \eta_V} \subseteq \mathfrak{m}_{\eta_V}^h \quad (h \in \mathbb{Z}_{>0}),$$

then

$$h^c H^{d-c} \cdot V \leq H^d.$$

In particular, if x is an isolated base point and every member has multiplicity at least a real number $r \geq 0$ at x , then $r^d \leq H^d$.

Proof. Intersect successively with general members of Λ . Before each next cut, remove the components already contained in the base locus. Each remaining component is cut properly, and the intersection cycle is effective. Its H -degree equals the H -degree of the preceding cycle; discarding effective components can only decrease this degree, since H

is nef. Starting with $[W]$, the H -degree of the codimension- c cycle is therefore at most H^d .

At η_V , the localized base ideal is primary for the maximal ideal. No discarded component of smaller codimension contains η_V , because V is an irreducible component of the base locus. The first c general equations $s_1, \dots, s_c$ form a parameter ideal I in the regular local ring $R = \mathcal{O}_{W, \eta_V}$. The coefficient of V in the intersection cycle is $\text{length}(R/I)$. Since $I \subseteq \mathfrak{m}^h$, monotonicity of Hilbert–Samuel multiplicity gives

$$\text{length}(R/I) = e(I, R) \geq e(\mathfrak{m}^h, R) = h^c.$$

Here the first equality uses that a parameter sequence in R is regular, and the last equality uses regularity of R . This proves the inequality. For an isolated point take $c = d$ and $h = \max\{1, \lceil r \rceil\}$. $\square$

Lemma 3.3 (Generic numerical subadjunction). *Let W be smooth projective, $\Theta \geq 0$ rational, and $V \subset W$ integral of dimension $f > 0$. Suppose (W, Θ) is lc at η_V and has an lc place centred on V . For a nef rational Cartier class P_V on V and a resolution $\rho : Y \rightarrow V$,*

$$K_Y(\rho^* P_V)^{f-1} \leq (K_W + \Theta)|_V P_V^{f-1}.$$

Proof. We give the numerical argument of [1, Lemma 7.1]. The dlt blowup [2, Theorem 2.10] gives

$$p^*(K_W + \Theta) = K_Q + B_Q + E_Q, \quad (Q, B_Q) \text{ dlt}, \quad E_Q \geq 0,$$

where B_Q is the truncated boundary and the effective surplus E_Q is absent near η_V . Choose a dlt stratum S minimal among those dominating V . Adjunction, followed by adding $E_Q|_S$, gives an effective rational divisor B_S and

$$K_S + B_S \sim_{\mathbb{Q}} (p|_S)^*((K_W + \Theta)|_V).$$

The stratum is not contained in E_Q . Minimality makes (S, B_S) klt over η_V , although it need not be globally lc.

Factor through the normalization and the connected-fibre base: $S \xrightarrow{q} V' \xrightarrow{\eta} V^\nu \rightarrow V$, with η finite. The discrepancy-round-up rank condition for q is one: over the generic point the effective klt boundary contributes zero round-up on strict divisors and allows poles only along exceptional divisors. Its pushforward to normal S is therefore $\mathcal{O}_S$, and connectedness gives rank one. Thus this is a klt-trivial fibration in [7, Definition 3.1]. Its formula is

$$\eta^*((K_W + \Theta)|_{V^\nu}) \sim_{\mathbb{Q}} K_{V'} + B_{V'} + M_{V'},$$

with b-nef moduli divisor M by [7, Theorem 3.5]. The discriminant on this initial base is effective: over a prime divisor, an inverse-image component of multiplicity $m \geq 1$ and boundary coefficient $b \geq 0$ bounds its lc threshold by $(1 - b)/m \leq 1$. A negative threshold still gives a positive discriminant coefficient. Only generic sub-klt singularities are required for this use of the formula.

Intersect with $f - 1$ copies of the pulled-back P_V . The effective discriminant has nonnegative intersection. So does the moduli trace, by pushing down from a higher base where it is nef. For compatible canonical Weil cycles,

$$\eta_* K_{V'} = (\deg \eta) K_{V^\nu} + \eta_* R_\eta, \quad R_\eta \geq 0.$$

Discard the nonnegative ramification term and divide by $\deg \eta$. Finally push down K_Y to V^ν ; the remaining exceptional cycles have zero testing intersection. This proves the inequality without requiring K_{V^ν} to be rational Cartier. $\square$

Lemma 3.4 (The canonical-intersection input). *Let W be smooth projective of any dimension, and let P_W be nef and rational Cartier. Suppose a positive-dimensional integral base component V of a subseries of $|kP_W|$ is isolated at its generic point, of codimension a , and its base ideal there is contained in $\mathfrak{m}^h$. Then on a resolution $\rho : Y \rightarrow V$,*

$$K_Y(\rho^*P_W)^{\dim V-1} \leq (K_W + cP_W)|_V P_W^{\dim V-1}, \quad 0 < c \leq ak/h.$$

Proof. The generic log canonical threshold of the base ideal satisfies $\lambda \leq a/h$, as seen by blowing up the smooth generic centre. On a log resolution, average sufficiently many general members of the subseries with total coefficient λ . This gives an actual effective boundary $\Theta \sim_{\mathbb{Q}} k\lambda P_W$ which is lc at the generic point of V and has an lc place centred on V . Lemma 3.3 gives the inequality with $c = k\lambda$. The construction uses actual general members of the given subseries, and requires lc only at η_V . Both the threshold bound and subadjunction are dimension-independent. $\square$

Lemma 3.5. *Let H be a Cartier divisor on X , let $\Lambda \subseteq |H|$ be a fixed nonempty linear system, and let $r \geq 0$ be real. If a Zariski dense set of points x admits $E_x \in \Lambda$ with $\text{mult}_x(E_x) > r$, then every point does.*

Proof. Use local parameters u, v, w and homogeneous parameters $[t_0 : \cdots : t_m]$ on the fixed linear system. Write the universal equation as $f = \sum_i t_i f_i$, and denote its coefficient functions by $c_{i,\alpha}(x)$, where $\alpha \in \mathbb{Z}_{\geq 0}^3$. The incidence Z is defined locally by

$$\sum_i t_i c_{i,\alpha}(x) = 0 \quad (|\alpha| \leq \lfloor r \rfloor).$$

These coefficients are the finite local jet coefficients. Thus Z is closed, and its proper projection has closed image. An image containing a Zariski dense subset equals X . $\square$

Lemma 3.6 (Full kernels and their actual fibres). *Let W be a smooth projective variety of dimension d , and let P_W be ample and rational Cartier. Fix kP_W Cartier and levels $\tau_0 < \cdots < \tau_d$ with common gap $\delta > 0$ and $k\delta \geq 4$. Suppose that, at a very general marked point z , all full spaces*

$$\mathcal{H}_i(z) = H^0(W, \mathcal{O}_W(kP_W) \otimes \mathfrak{m}_z^{\lceil k\tau_i \rceil})$$

are nonzero, and the lowest base locus has a positive-dimensional component through z . Then there is an algebraic covering family of proper integral components V_t , common to two adjacent base loci, with $1 \leq f = \dim V_t < d$. At a general incidence point the entire higher-kernel base ideal satisfies

$$\mathfrak{b}_t \subset I_{V_t}^h, \quad h = \lceil k\delta/2 \rceil, \quad k/h \leq 2/\delta.$$

Consequently, on a resolution V_t^ ,*

$$(3) \quad \begin{aligned} 0 < P_W^f V_t &\leq (2/\delta)^{d-f} P_W^d, \\ K_{V_t^*} P_W^{f-1} &\leq (K_W + cP_W)|_{V_t} P_W^{f-1}, \quad 0 < c \leq 2(d-f)/\delta. \end{aligned}$$

For a morphism $g : W \rightarrow B$, take the reduced component F through a general incidence point of a fibre of $V_t \rightarrow g(V_t)$. If F is proper in its smooth integral projective ambient fibre W_b , the restricted series is nonzero, F is an isolated base component, and its ideal is contained in I_F^h . The same bounds apply on W_b , using its dimension and canonical divisor.

Proof. This is the smooth-ambient form of [1, Lemmas 7.3–7.4], also used in [3, Lemmas 4.5–4.6]. Here are the local details needed for its two applications.

On the open where the finitely many jet evaluation maps have constant rank, their full kernels are vector bundles. Their base loci are nested. Track a component through the mark across the $d + 1$ levels. Its dimension belongs to $\{1, \dots, d - 1\}$, so two adjacent tracked components coincide. A finite extension of the parameter field defines the chosen geometric components; spreading and shrinking gives an integral parameter variety T and an incidence $\Gamma \subset T \times W$ dominating W .

A frame of the higher kernel has the form $s(t, w) = \sum_{\alpha} a_{\alpha}(t)s_{\alpha}(w)$ for fixed global sections s_{α} . A parameter derivative of order $j < h$ is still global and loses at most j orders at the moving mark. Since

$$\lceil k\tau_{i+1} \rceil - \lceil k\tau_i \rceil \geq k\delta - 1 \geq h - 1,$$

each such derivative belongs to the full lower kernel, hence vanishes on Γ . At a general smooth point, dominance of $\Gamma \rightarrow W$ makes parameter directions span its normal space. Choose transverse parameter coordinates t' and write the incidence locally as $u = t' - F(t'', w) = 0$. With t'', w fixed, differentiation in t' is differentiation in u to every order. The preceding vanishing therefore says $s \in I_{\Gamma}^h$. Characteristic zero and faithful flatness of the analytic local ring give the same ordinary ideal containment algebraically. Restrict the finite frame to a general parameter fibre to obtain $\mathfrak{b}_t \subset I_{V_t}^h$. The degree bound follows from Lemma 3.2; the canonical bound follows from Lemma 3.4.

For the restriction assertion, remove the other base components and choose the point where $V_t \rightarrow g(V_t)$ is smooth over its *actual image*. Its scheme fibre is reduced there, so

$$I_{V_t} \mathcal{O}_{W_b} = I_F, \quad I_{V_t}^h \mathcal{O}_{W_b} = I_F^h.$$

The base support locally becomes exactly F . Since F is proper, the restricted series cannot vanish identically on W_b ; its localized base ideal is primary at η_F . Apply the preceding degree and canonical lemmas to these actual restricted sections. $\square$

Proposition 3.7. *Let P be an ample rational Cartier divisor on X , and put $\rho = (P^3)^{1/3}$. Suppose that, for some positive integer k_* with k_*P Cartier, a Zariski dense set of points x admits $E_x \in |k_*P|$ with $\text{mult}_x(E_x) > 4k_*\rho$. Then there is a covering family of integral subvarieties V of one fixed dimension $d \in \{1, 2\}$ such that*

$$0 < P^d \cdot V \leq 3^{3-d}(P^3)^{d/3}.$$

Moreover, on a resolution $Y \rightarrow V$, with P_Y the pullback,

$$(4) \quad K_Y P_Y^{d-1} \leq \frac{3(3-d)}{\rho} P_Y^d.$$

The marked incidence dominates X .

Proof. Lemma 3.5 makes the multiplicity condition valid at every point. Taking powers of the defining sections allows k to be arbitrarily large and divisible by k_* . Put

$$\delta = \frac{2\rho}{3}, \quad \tau_i = \frac{3}{2}\rho + i\delta \quad (0 \leq i \leq 3), \quad q_i = \lceil k\tau_i \rceil.$$

Choose k so that $k\delta \geq 4$ and $k\rho/2 > 1$. All full kernel spaces

$$V_i(x) = H^0(X, \mathcal{O}_X(kP) \otimes \mathfrak{m}_x^{q_i})$$

are nonzero, because $\tau_3 = 7\rho/2 < 4\rho$. Write $B_i(x)$ for their base loci. They are proper and nested. The point x cannot be isolated in $B_0(x)$: otherwise Lemma 3.2 would give $q_0^3 \leq k^3 P^3$, whereas $q_0 \geq 3k\rho/2$.

Apply Lemma 3.6 with $W = X$ and these four levels. It gives a common component V_t of dimension $d \in \{1, 2\}$ and a covering incidence, together with ordinary-power containment for the entire higher base ideal, where $h = \lceil k\delta/2 \rceil$.

The higher base ideal is primary there, since V_t is its component. Lemma 3.2, applied to $H = kP$, gives

$$h^{3-d}k^dP^d \cdot V_t \leq k^3P^3, \quad P^d \cdot V_t \leq (k/h)^{3-d}P^3 \leq (3/\rho)^{3-d}P^3.$$

Lemma 3.4 gives $K_Y P_Y^{d-1} \leq (3-d)(k/h)P_Y^d$, because $K_X \sim 0$. Since $k/h \leq 3/\rho$, this is (4). Ampleness gives positivity, and the incidence is dominant. $\square$

In dimension three the alternatives are a curve with $P \cdot C \leq 9(P^3)^{1/3}$, or a surface with $P^2 \cdot V \leq 3(P^3)^{2/3}$. The canonical-intersection bound is what permits the surface alternative to be treated in the following subsection.

3.1. The scalar bound and its movable witness.

Lemma 3.8 (Bounded-degree obstruction). *Let Y be an auxiliary smooth projective variety of dimension $f \in \{1, 2, 3\}$. Let B_0 be a reduced simple normal crossing divisor or zero, L a nef Cartier divisor of full nef dimension, and P_Y a nef rational Cartier divisor. If*

$$0 < P_Y^f \leq C_0, \quad P_Y - rL \text{ is nef}, \quad K_Y + B_0 + L \text{ is big},$$

and

$$(K_Y + B_0 + L)P_Y^{f-1} \leq C_1 P_Y^f,$$

then

$$r \leq (m_f C_1)^{f-1} C_0,$$

where m_f depends only on f .

Proof. Birkar–Zhang [8, Theorem 1.3], with boundary coefficients in $\{0, 1\}$ and Cartier index one for L , makes $|m_f(K_Y + B_0 + L)|$ birational. Resolve its base ideal and let H be the free moving part. Then H is nef and big and

$$H P_Y^{f-1} \leq m_f C_1 P_Y^f.$$

Khovanskii–Teissier gives

$$P_Y H^{f-1} \leq P_Y^f \left(\frac{H P_Y^{f-1}}{P_Y^f} \right)^{f-1} \leq (m_f C_1)^{f-1} C_0.$$

General hyperplanes through the image of a very general point under the birational morphism defined by H cut an effective curve cycle containing a curve C through that point. Full nef dimension and integrality give $LC \geq 1$. Thus $r \leq P_Y C \leq P_Y H^{f-1}$. For $f = 1$ take $C = Y$. This is [3, Lemma 4.1]; in particular its constants are independent of the varying auxiliary variety. $\square$

Proposition 3.9. *Assume $\kappa(X, D) = 3$ and $D^3 = 0$. With $P_t = r_t(D + tA)$ as in Lemma 3.1, put $\rho_t = (P_t^3)^{1/3}$. For all sufficiently small terms of the fixed rational sequence, and at a very general point x , every member $E \in |kP_t|$, for every positive integer k with kP_t Cartier, satisfies*

$$\text{mult}_x(E) \leq 4k\rho_t.$$

Proof. Suppose violations persist as $t \rightarrow 0$. For fixed t, k , the high-multiplicity condition is closed in the incidence space, so a violation at a sufficiently very general point yields a dense set of violations. Use Proposition 3.7, together with its canonical bound (4), and write $Y \rightarrow V$ for the resolution of a general member. For $f = \dim V \in \{1, 2\}$ it gives

$$0 < P_Y^f \leq 3^{3-f} \rho_t^f, \quad K_Y P_Y^{f-1} \leq \frac{3(3-f)}{\rho_t} P_Y^f.$$

The volume window $P_t^3 \rightarrow \epsilon^3 > 0$ makes both bounds uniform in t . Let L be the pullback of D ; then $P_Y - r_t L$ is nef.

Choose the incidence point above the very general full-nef locus and outside the images of all families of rational curves. The latter exclusion is possible because $K_X \sim 0$ makes X non-uniruled. Since $Y \rightarrow V \subset X$ is generically finite, Y is non-uniruled and L has full nef dimension; this is the very-general-images argument of [3, Lemma 4.2]. By [14], K_Y is pseudo-effective, and Lemma 2.1 makes $K_Y + L$ big. Also $LP_Y^{f-1} \leq P_Y^f/r_t$. Lemma 3.8 now bounds r_t uniformly, contrary to $r_t \rightarrow \infty$. There are only countably many t, k and jet orders, so one very general locus handles all the stated section spaces. $\square$

The essential surface input is the bound for $K_Y P_Y$. For $\nu(D) = 2$, a degree bound alone gives $D^2 V = 0$ and does not give $DAV = 0$. The canonical bound and the preceding bounded-degree obstruction supply the missing implication. Neither case $\nu(D) = 1$ or 2 uses a condition on $D \cdot c_2(X)$.

Proposition 3.10. *Assume the scalar bound at a very general point x , and fix small t with $\rho_t < 2\epsilon$ and $r_t > Q$. On the blowup $\pi : \widehat{X} \rightarrow X$ at x there is an actual strongly movable class $\gamma = f_*(A_1 A_2)$ such that*

$$J\gamma > 0, \quad \pi^*(P + \lambda D)\gamma < 40\epsilon J\gamma \quad (0 \leq \lambda \leq Q).$$

Here J is exceptional, f is one smooth birational model, and the A_i are ample integral Cartier divisors.

Proof. The class $2\pi^*P - 40\epsilon J$ is not pseudo-effective. Otherwise its convex combination with the big class $2\pi^*P$ would make $2\pi^*P - cJ$ big for rational $8\rho_t < c < 40\epsilon$. An effective divisor in a divisible multiple of this class would push down to a member of $|2kP|$ with multiplicity exceeding the bound in Proposition 3.9. Movable-curve duality [14] gives a generating class $\gamma = f_*(A_1 A_2)$ with

$$2\pi^*P\gamma < 40\epsilon J\gamma.$$

Strict negativity permits rational approximation of the ample classes and clearing denominators. Nefness gives $J\gamma > 0$. For $0 \leq \lambda \leq Q$ the divisor

$$2P - (P + \lambda D) = (r_t - \lambda)D + r_t t A$$

is nef, which gives the required inequalities. Fix this actual model and complete intersection before reducing characteristic. $\square$

Lemma 3.11 (A positive-Iitaka boundary). *Suppose $n(X, D) = 3$. If $G \geq 0$ is rational, (X, G) is klt, and $\kappa(G) > 0$, then $G + cD$ is big for every rational $c > 0$.*

Proof. We spell out [3, Proposition 3.3] in the only threefold instance required below. Run the ordinary G -MMP to a good model; ordinary threefold abundance is [9]. Its adjoint $U = K_{Y_0} + G_{Y_0}$ is semiample. Let N_0 be the trace of the fixed nef b-divisor determined by D on X . For small $s > 0$, this finite first program is also negative for $G + sD$, and $(Y_0, G_{Y_0} + sD)$ is generalized klt.

Choose dU Cartier and free, and choose $0 < s_0 < s$ such that

$$(1 - s_0/s)/d > 12s_0/s.$$

Run a generalized $(U + s_0 N_0)$ -program. The nef data are rational and NQC; existence of the steps follows by ordinary klt approximation, and termination is [10, Theorem 1.2]. For a negative ray the generalized length bound supplies a generator C with $(U + sN_0)C \geq -12$; see [3, Lemma 3.1]. If $UC > 0$, Cartier integrality gives $UC \geq 1/d$, whereas

$$(U + s_0 N_0)C \geq (1 - s_0/s)/d - 12s_0/s > 0,$$

a contradiction. Every step is therefore U -trivial. The free line bundle dU descends through each contraction and pulls back through its flip. On the final model Y , U remains semiample, $Q_0 = U + s_0N$ is nef, and a common resolution gives

$$(5) \quad p^*G = q^*U + E_0, \quad p^*(G + s_0D) = q^*Q_0 + E_1, \quad E_0, E_1 \geq 0 \text{ } q\text{-exceptional.}$$

Moreover Q_0 has full nef dimension. Otherwise a very general compact nef-reduction fibre has a curve through its very general point avoiding $q(\text{Exc } q)$, which has codimension at least two. Nefness of U , pseudo-effectivity of N , and generic nonnegativity of pseudo-effective degrees give $UC = NC = 0$. Lifting this curve in both comparisons gives D -degree zero through a very general point of X , a contradiction.

The contraction $h : Y \rightarrow S$ defined by U has $\dim S = \kappa(G) > 0$ and $U \sim_{\mathbb{Q}} h^*A_S$ with A_S ample. On a very general positive-dimensional fibre F ,

$$K_F + G_Y|_F \sim_{\mathbb{Q}} 0, \quad Q_0|_F = s_0N|_F \text{ is nef of full nef dimension.}$$

Its dimension is at most two, so Lemma 2.1 makes $Q_0|_F$ big. For zero-dimensional fibres relative bigness is automatic. Relative bigness follows on the generic fibre: for a fixed ample H , one of the countably many closed section loci for $(\ell Q_0 - H)|_{F_s}$ contains a very general point, hence is the whole shrunken flat base. Adding a sufficiently large multiple bU now makes $Q_0 + bU$ big globally. The comparisons imply that $(1 + b)G + s_0D$ is big. Finally, for $0 < \lambda < \min\{1/(1 + b), c/s_0\}$,

$$G + cD = \lambda((1 + b)G + s_0D) + (1 - \lambda(1 + b))G + (c - \lambda s_0)D$$

is big. The higher-dimensional version is the same argument with lower-dimensional adjoint bigness and terminating programs as inputs. Here every program has dimension three and every fibre input has dimension at most two. $\square$

Completion of the proof of Lemma 1.1. Proposition 3.9 gives the multiplicity bound. For fixed Q , take t smaller so that $\rho_t < 2\epsilon$ and $r_t > Q$; Proposition 3.10 then gives the movable witness. $\square$

4. PROOF OF THE SEVENFOLD JET LEMMA

We prove Lemma 1.2. The auxiliary results below exclude the possible moving base components; Proposition 4.5 gives the asserted jet system.

For $\nu(D) = 1$ or 2 , fix the normalization $P = P_t$, and let $Q > 0$ be an integer. Subscripts denote pullbacks from the factors of $X \times X$. Put

$$Z = \mathbb{P}(\mathcal{O}_X(D)_1 \oplus \mathcal{O}_X(D)_2), \quad \xi = c_1(\mathcal{O}_Z(1)), \quad R = P_1 + P_2 + Q\xi.$$

We use the convention $\pi_*\mathcal{O}_Z(a) = \text{Sym}^a(\mathcal{O}_X(D)_1 \oplus \mathcal{O}_X(D)_2)$. Thus $\dim Z = 7$ and R is ample. The complement of the two summand sections is the torus open. On a slice over one fixed base coordinate,

$$Z_{\text{sl}} = \mathbb{P}(\mathcal{O}_X \oplus \mathcal{O}_X(D)), \quad R_{\text{sl}} = P + Q\xi, \quad \dim Z_{\text{sl}} = 4.$$

Lemma 4.1. *With $P_a = P + aD$, the projective-bundle formulas give*

$$R^7 = 140 \int_0^Q P_a^3 P_{Q-a}^3 da, \quad R_{\text{sl}}^4 = 4 \int_0^Q P_a^3 da.$$

For fixed Q , these tend to $140Q\epsilon^6$ and $4Q\epsilon^3$, respectively, as $t \rightarrow 0$. Also

$$K_Z \sim D_1 + D_2 - 2\xi, \quad K_{Z_{\text{sl}}} \sim D - 2\xi.$$

Proof. For a split rank-two bundle, $\pi_*(\xi^j) = \sum_{a+b=j-1} D_1^a D_2^b$ for $j \geq 1$. Expand R^7 and integrate the binomial terms to obtain the first identity; its coefficient is $7!/(3!3!) = 140$. The analogous one-factor expansion gives the second identity. Since $D^3 = 0$, $P^3 \rightarrow \epsilon^3$, and $r_t \rightarrow \infty$, $(P + aD)^3/P^3 \rightarrow 1$ uniformly for $0 \leq a \leq Q$. The two limits follow. The canonical formulas are the rank-two projective-bundle formula with $K_X \sim 0$. $\square$

Lemma 4.2 (The axes of a multisection). *Assume $n(X, D) = 3$. Let $F \subset \mathbb{P}(\mathcal{O}_X \oplus \mathcal{O}_X(D))$ be an integral multisection of degree $e > 0$ meeting the torus open. Its two axis intersections push down to effective integral divisors G_1, G_2 on X satisfying*

$$(6) \quad G_1 - G_2 \sim \pm eD, \quad QG_i P^2 \leq R_{\text{sl}}^3 F.$$

For $G = G_1 + G_2$ and every sufficiently small rational $b > 0$, (X, bG) is klt, $\kappa(bG) > 0$, and $bG + D$ is big.

Proof. The norm of the ratio of the two fibre coordinates gives the linear equivalence. If S is either axis, $R_{\text{sl}} - QS$ and $R_{\text{sl}} - P$ are nef. Intersecting these classes against the effective cycle $S \cdot F$ gives the degree inequalities.

If $\kappa(bG) = 0$, ordinary threefold abundance gives a good $(K_X + bG)$ -model Y with $K_Y + bG_Y \sim_{\mathbb{Q}} 0$. The transform of K_X is linearly trivial, so $G_Y = 0$: every component of G has been contracted. On a common resolution $a : U \rightarrow X$, $c : U \rightarrow Y$, the signed relation in (6) expresses $a^*D \sim_{\mathbb{Q}} E$, with E c -exceptional. Divisors exceptional over X are also exceptional over Y , since this MMP extracts none. Since a^*D is nef, negativity gives $E \leq 0$. Its class is pseudo-effective, so intersection with an ample square forces $E = 0$. This contradicts full nef dimension. Thus $\kappa(bG) > 0$, and Lemma 3.11 gives bigness. The axis construction and negativity argument are dimension-independent; only the ordinary good-model input is being specialized to threefolds. $\square$

Lemma 4.3 (A dominant family of fixed-cover graphs). *Let T_1, T_2 be fixed non-uniruled integral projective varieties over $\mathbb{C}$ of the same dimension. If an integral algebraic family of graphs of birational maps $T_1 \dashrightarrow T_2$ dominates $T_1 \times T_2$, both varieties are birational to an abelian variety.*

Proof. Identify the two varieties by one map in the family. Hanamura's theorem, in the form [13, Theorem 3.8], supplies a smooth projective birational model Y with $\text{Bir}(Y)_{\text{red}}$ a group scheme locally of finite type. Its identity component is $\text{Aut}^0(Y)$, an abelian variety acting regularly.

Conjugate the family to Y , take the closure of its relative graphs, and shrink the integral parameter space so that these graphs are flat and geometrically integral, with both projections birational. This shrinking is necessary: graph flatness is not preserved automatically by birational conjugation [13, Remark 3.6]. The flat-graph representability theorem [13, Proposition 3.7] then gives a morphism from the reduced parameter variety to $\text{Bir}(Y)_{\text{red}}$. Its connected image lies in one translate of the identity component. After translating by a fixed member, product dominance implies that a general point has a dense orbit under that abelian group. Such an orbit is the quotient by its closed stabilizer, hence an abelian variety. It is therefore a birational model of Y . No canonical nonvanishing theorem on T_i is used. $\square$

Lemma 4.4 (Returning from an abelian model). *Let $p : Y \rightarrow A_0$ be a birational morphism from a smooth projective variety to an abelian variety. If L is nef Cartier of full nef dimension on Y , then $K_Y + L$ is big.*

Proof. The numerical pushforward $N = p_*L$ is pseudo-effective on A_0 , thus nef. If it were not big, the kernel of the semipositive Hermitian form of its integral class would

define a positive-dimensional abelian subvariety on whose translates N is numerically trivial. The image of the exceptional locus of p has codimension at least two. A general translate meets that image in codimension at least two; complete-intersection curves in the translate can be chosen through a very general point avoiding it. Their strict lifts have L -degree zero, a contradiction. Thus N is big.

Negativity gives $p^*N - L = E_0 \geq 0$, with E_0 exceptional. Since A_0 is smooth and $K_{A_0} \sim 0$, $K_Y \sim E_K \geq 0$, with positive coefficient along every p -exceptional divisor. Choose $0 < u < 1$ rational with $E_K - uE_0 \geq 0$. Then

$$K_Y + uL \sim_{\mathbb{Q}} up^*N + (E_K - uE_0)$$

is big. Adding $(1-u)L$ proves the assertion. This lemma has no dimension restriction. $\square$

Proposition 4.5 (The two-slot jet system). *Assume $n(X, D) = 3$ and $D^3 = 0$. For a suitable fixed integer $Q > 0$ and all sufficiently small t , there is a point z in the torus open of Z such that $\varepsilon(R; z) > 8$. Consequently, for one sufficiently divisible integer $k_0 > 0$ with k_0P Cartier, finitely many sections of k_0R generate jets through order $8k_0$ at z .*

Proof. We write out the dimension-three instance of [3, Proposition 5.3]. All constants below are fixed before $t \rightarrow 0$. Shrink the normalization sequence so that

$$\epsilon^3/2 < P^3 < 2\epsilon^3.$$

Choose $\delta = 8\epsilon$, and then choose Q so large that the eight levels $\tau_i = 9 + i\delta$, $0 \leq i \leq 7$, satisfy $\tau_7^7 < 70Q\epsilon^6$. By Lemma 4.1, for small t ,

$$\tau_7^7 < R^7 < 280Q\epsilon^6, \quad R_{\text{sl}}^4 < 8Q\epsilon^3, \quad (2/\delta)^3 8\epsilon^3 = 1/8 < 1.$$

Suppose $\varepsilon(R; z) \leq 8$ throughout the torus open for arbitrarily small t . Asymptotic Riemann–Roch and the jet count make all eight full kernels nonzero for sufficiently large divisible k . Through every very general z there is a curve with R -degree divided by multiplicity less than τ_0 . Every section in the lowest kernel vanishes identically on that curve, by the intersection multiplicity inequality. Hence its base locus has a positive-dimensional component through z . Lemma 3.6 gives a covering family of integral proper components $V \subset Z$, with $f = \dim V$, satisfying

$$(7) \quad \begin{aligned} 0 < R^f V &\leq (2/\delta)^{7-f} R^7, \\ K_{V^*} R^{f-1} &\leq c_f R^f V, \quad c_f = 2(7-f)/\delta. \end{aligned}$$

Indeed K_Z is minus the axes, and V meets their complement, so its ambient canonical contribution is nonpositive.

The slice fibres. For either projection, take a general fibre component F of $V \rightarrow \text{im}(V \rightarrow X)$ through a general incidence point whose two images are sufficiently very general on X . If $1 \leq s = \dim F \leq 3$, its ambient fibre is the smooth fourfold Z_{sl} . Lemma 3.6 applies to the actual restricted series, giving

$$(8) \quad \begin{aligned} 0 < R_{\text{sl}}^s F &\leq (2/\delta)^{4-s} R_{\text{sl}}^4, \\ K_{F^*} R_{\text{sl}}^{s-1} &\leq (K_{Z_{\text{sl}}} + c_s R_{\text{sl}})|_F R_{\text{sl}}^{s-1}, \quad c_s \leq 2(4-s)/\delta. \end{aligned}$$

Let $W_0 \subset X$ be its image in the other factor.

If $F \rightarrow W_0$ is generically finite and $\dim W_0 < 3$, the pullback L of D is nef Cartier of full nef dimension on F^* , and K_{F^*} is pseudo-effective by the very-general-images argument. Lemma 2.1 gives $K_{F^*} + L$ big. The testing class $P_Y = R_{\text{sl}}|_{F^*}$ satisfies $P_Y - r_t L$ nef, and (8) gives uniform volume and canonical ratios. Lemma 3.8 bounds r_t , a contradiction.

If $F \rightarrow W_0$ has one-dimensional fibres, F is the entire projective-line bundle over W_0 . Put $w = \dim W_0 = s - 1$. For $w = 0$, (8) gives

$$Q = R_{\text{sl}} F \leq (2/\delta)^3 R_{\text{sl}}^4 < Q.$$

For $w = 1, 2$, resolve W_0 and take its induced bundle $p_0 : F^* \rightarrow W_0^*$. Cancelling the relative canonical term $-2\xi + p_0^*D$, which equals the ambient canonical restriction, gives

$$p_0^*K_{W_0^*}R_{\text{sl}}^w \leq c_s R_{\text{sl}}^{w+1}F.$$

Its term linear in ξ is $wQK_{W_0^*}P^{w-1}$; all other terms are nonnegative because $K_{W_0^*}$ is pseudo-effective and P, D are nef. For $r_t \geq Q$, expansion gives

$$(9) \quad \begin{aligned} (w+1)QP^wW_0 &\leq R_{\text{sl}}^{w+1}F \leq 7QP^wW_0, \\ K_{W_0^*}P^{w-1} &\leq (7c_s/w)P^wW_0. \end{aligned}$$

The first lower bound and (8) bound P^wW_0 uniformly. Applying Lemmas 2.1 and 3.8 on W_0^* is again contradictory.

The remaining slice case is a degree- e multisection over X . Lemma 4.2 gives $G = G_1 + G_2$, $bG + D$ big, and $GP^2 \leq 2R_{\text{sl}}^3F/Q$. On a log resolution $a : Y \rightarrow X$, let B_0 be the reduced strict support of G together with every exceptional divisor. Decrease b so that

$$K_Y + B_0 + a^*D \sim_{\mathbb{Q}} a^*(bG + D) + E_b, \quad E_b \geq 0.$$

This adjoint is big. Testing with $P_Y = a^*P$, all exceptional divisors contribute zero, and

$$(K_Y + B_0 + a^*D)P_Y^2 \leq GP^2 + DP^2 \leq C, \quad DP^2 \leq P^3/r_t.$$

The positive lower bound on P^3 gives a uniform adjoint ratio. Lemma 3.8, with this reduced boundary, excludes the case. Thus neither projection has fibre dimension 1, 2, or 3.

A fibre of dimension four would make V the full inverse image of its image W_0 in that factor. Since V is proper, $\dim W_0 \leq 2$; the other projection would then have fibre dimension $\dim W_0 + 1 \in \{1, 2, 3\}$, already excluded. Both projections are therefore generically finite and $f \leq 3$. For $f < 3$, (7) and the lower-dimensional bounded-degree obstruction exclude V as before. Only threefold correspondences dominant over both copies of X remain.

Branch divisors and fixed finite covers. Set $C_0 = (2/\delta)^4 280Q\epsilon^6$. On a simultaneous resolution Y of a general correspondence, with projections $g_i : Y \rightarrow X$ and testing class $P_Y = R|_Y$,

$$(10) \quad 0 < P_Y^3 \leq C_0, \quad K_Y P_Y^2 \leq (8/\delta)P_Y^3, \quad P_Y - r_t g_i^*D \text{ is nef.}$$

Also $R - P_i$ is nef, so the projection degrees satisfy $\deg g_i \leq C_0/P^3 \leq 2C_0/\epsilon^3$. Resolve and shrink the algebraic parameter family so its fibres are smooth and geometrically integral. After a finite parameter extension, spread out its finitely many generic Jacobian components.

Suppose that the images of one branch-divisor component sweep X , and let $J \subset X$ be a general such image. Pulling back a nowhere-vanishing three-form from X gives an effective canonical divisor on Y , containing a ramification divisor over J with positive integral coefficient. Thus

$$0 < d_J := P^2J \leq K_Y P_Y^2 \leq (8/\delta)C_0.$$

Generic numerical subadjunction for (X, J) and Hodge index give on a resolution J^*

$$(11) \quad K_{J^*}P \leq J^2P \leq \frac{(JP^2)^2}{P^3} \leq C_2 d_J, \quad C_2 = 16C_0/(\delta\epsilon^3).$$

Here (X, J) is lc at η_J ; global lc is unnecessary. Hodge index applies to J without assuming J nef. Since the images sweep, J^* is non-uniruled and the pulled-back D has full nef dimension. The testing volume is d_J , bounded above, its canonical ratio is bounded by C_2 , and $P - r_t D$ is nef. Lemmas 2.1 and 3.8 give a uniform bound for r_t . Hence sweeping branch divisors are impossible for small t .

For this fixed t , retain the relative Jacobian components whose general fibre images are divisors on X . Remove their finitely many proper image closures, obtaining a fixed smooth open U . Every finite normal Stein cover of a general g_i is unramified at the codimension-one points of U ; purity makes it finite étale over U . The fundamental group of U is finitely generated, so it has finitely many subgroups of the bounded indices in question. Normal covers of X are determined by their function fields. There are consequently only finitely many possible Stein covers in each slot, as in [1, Lemma 7.5]. Degree bounds alone would not imply this.

The abelian model and the final exclusion. Each Y is birational to both of its Stein covers. Fix a pair T_1, T_2 occurring among the original members satisfying (10). The projected incidence dominates $X \times X$. The possible graph families for the finitely many pairs belong to countably many Hilbert schemes; stratify for flat integral fibres and birational projections. Uncountability of $\mathbb{C}$ implies that one such integral graph family has evaluation dominant over $X \times X$. Its evaluation is also dominant on $T_1 \times T_2$: this product is integral, finite over $X \times X$, and of the same dimension.

The covers are non-uniruled, since they are finite over X . Lemma 4.3 makes them birational to an abelian variety A_0 . Return to an *original* correspondence for this pair and resolve further to get a birational morphism $p : Y \rightarrow A_0$. This preserves the numerical bounds (10); additional exceptional canonical divisors have zero intersection with the pulled-back testing square. The divisor $L = g_1^*D$ remains nef Cartier of full nef dimension. Lemma 4.4 gives $K_Y + L$ big, while

$$(K_Y + L)P_Y^2 \leq (8/\delta + 1/r_t)P_Y^3.$$

Lemma 3.8 now excludes this final case. The cover lists and graph families may depend on t ; their sizes enter none of the uniform numerical bounds. Failures therefore cannot persist along $t \rightarrow 0$.

Finally, choose rational $8 < a < \varepsilon(R; z)$. On the point blowup $b : \tilde{Z} \rightarrow Z$, $b^*R - aJ_z$ is ample. For sufficiently divisible k_0 , Serre vanishing makes restriction surjective modulo $\mathfrak{m}_z^{k_0 a}$, and $k_0 a \geq 8k_0 + 1$. These are the required order- $8k_0$ jets. $\square$

5. PROOF OF THE DETERMINANT COMPARISON LEMMA

We prove Lemma 1.3 by comparing upper and lower multiplicity bounds for the same determinant divisor.

All characteristic-zero data are fixed before reduction. Frobenius is used only on smooth positive-characteristic fibres; the final conclusion concerns $X/\mathbb{C}$.

Lemma 5.1. *If sections of a line bundle L on Z generate jets through order j at z , then products of N such sections generate jets through order Nj for L^N .*

Proof. Every monomial of degree at most Nj is a product of N monomials of degrees at most j . Lift the latter to sections with those leading terms and multiply. The product has the desired leading term and only higher-order errors. In a basis ordered by total degree this gives a triangular matrix with nonzero diagonal on the finite jet quotient, hence surjectivity. $\square$

Proposition 5.2. *Let Y be a smooth positive-characteristic fibre of a model of X , and use the same notation for the specialized divisor data, over an algebraically closed field of characteristic p . On $\mathbb{P}(\mathcal{O}_Y(D)_1 \oplus \mathcal{O}_Y(D)_2)$, suppose a fixed finite system of sections of $k_0(P_1 + P_2 + Q\xi)$ generates jets through order $8k_0$ at a point z off the axes, where k_0P is Cartier. Choose T so that all $T + aD$ for $0 \leq a \leq (k_0 - 1)Q$ have sections nonzero at the two base points. Put $N_p = \lfloor p/k_0 \rfloor$, $B_p = k_0N_pP + T$. For $p \geq 8k_0$, the two-slot evaluation map has generic rank p^6 .*

Proof. Lemma 5.1 gives order $8k_0N_p$ for N_pk_0R . Put $d_p = (p - k_0N_p)Q$. A weight-zero section of $d_p\xi + T_1 + T_2$ nonzero at z pads this system to $pQ\xi + B_{p,1} + B_{p,2}$. Its multiplication is invertible on the local jet quotients. Moreover

$$8k_0N_p \geq 8(p - k_0 + 1) \geq 7(p - 1).$$

In seven regular parameters, a monomial surviving modulo $(u_1^p, \dots, u_7^p)$ has degree at most $7(p - 1)$. Thus the padded sections surject onto this Frobenius quotient.

Let $F : Y \rightarrow Y'$ be relative Frobenius, let D' be the twist, and put $\mathcal{E} = F_*\mathcal{O}_Y(B_p)$. Smoothness makes F finite flat of degree p^3 , so $\text{rk } \mathcal{E} = p^3$. The tensor product A_* of the two local Frobenius-fibre algebras has dimension p^6 . If t is the fibre coordinate and $t(z) = c \neq 0$, the full quotient is $A_*[t]/(t^p - c^p)$, free over A_* with basis $1, t, \dots, t^{p-1}$. Projection to the coefficient of 1 keeps exactly weights $j = pa$, with $0 \leq a \leq Q$. Consequently the natural map

$$\begin{aligned} \bigoplus_{a+b=Q} H^0(Y, B_p + paD) \otimes H^0(Y, B_p + pbD) \\ \otimes \mathcal{O}_{Y' \times Y'}(-aD'_1 - bD'_2) \longrightarrow \mathcal{E} \boxtimes \mathcal{E} \end{aligned}$$

spans A_* at the base pair. The projection formula identifies these summands with the evaluation map; the surviving fibre scalars c^{pa} are nonzero. The map therefore has generic rank p^6 . No reducedness assumption on A_* is used. $\square$

The displayed direct sum is over nonnegative integers a, b . For a characteristic-zero application the finite jet system, point, padding twists, and divisor data must be fixed first, then spread out. Surjectivity of a fixed finite evaluation matrix persists on a nonempty model open. Frobenius is applied only to its positive-characteristic fibres, not to $X/\mathbb{C}$.

Proposition 5.3 (The finite-data contradiction). *Suppose the following data have been fixed over $\mathbb{C}$: the jet system in Proposition 4.5; a smooth complete intersection flag with successive divisor classes $h_i = l_i P$ for $i = 1, 2$ and final curve C , such that $\mu_{\min}(\Omega_X^1|_C) \geq 0$; and the point blowup $\pi : \widehat{X} \rightarrow X$ at x , with exceptional divisor J , together with an actual strongly movable class $\gamma = f_*(A_1 A_2)$, where the A_i are ample Cartier divisors on one smooth birational model, satisfying*

$$J\gamma > 0, \quad \pi^*(P + \lambda D)\gamma \leq 40\epsilon J\gamma \quad (0 \leq \lambda \leq Q).$$

If

$$P^3 < (2\epsilon)^3, \quad (2P + QD)P^2 < 4P^3, \quad (14\epsilon)^3 < \frac{1}{4}, \quad 80\epsilon < \frac{3}{4},$$

these data cannot coexist.

Proof. This is [3, Theorem 6.1] with its divisors $(D_{\text{source}}, H, S) = (P, P, D)$ and $K_X \sim 0$. We give the rank and vanishing estimates that close the application in this setting.

Fixed data and generic rank. Choose the padding divisor T and the finitely many nonvanishing padding sections as in Proposition 5.2. Spread out the flag, canonical trivialization, jet sections, blowup, and the actual complete intersection witness for γ . Shrinking the model preserves smoothness, ampleness, the finite jet surjection, and all intersection numbers. Since $\deg(\Omega_X^1|_C) = K_X C = 0$, the least-slope condition makes this curve bundle semistable of slope zero. Openness of semistability preserves it after further shrinking. There remain closed fibres in arbitrarily large characteristics. On such a smooth fibre Y , with relative Frobenius $F : Y \rightarrow Y'$, set

$$N_p = \lfloor p/k_0 \rfloor, \quad B_p = k_0 N_p P + T, \quad \mathcal{E}_p = F_*\mathcal{O}_Y(B_p), \quad s_p = p^3, \quad R_p = s_p^2 = p^6.$$

Proposition 5.2 gives generic evaluation rank R_p . Choose R_p generically independent evaluation columns of weights (a, b) , $a + b = Q$. Their determinant is a nonzero section

$$\sigma \in H^0(Y' \times Y', \mathcal{Q}_1 \boxtimes \mathcal{Q}_2), \quad \mathcal{Q}_i = (\det \mathcal{E}_p)^{s_p} \otimes \mathcal{O}_{Y'}(m_i D'),$$

where $m_i \geq 0$ and $m_1 + m_2 = QR_p$. Let E_σ be the effective divisor of σ , so

$$E_\sigma \in |\mathcal{Q}_1 \boxtimes \mathcal{Q}_2|.$$

Rank on the diagonal. Let $\Delta_F = Y \times_{Y'} Y$, and put

$$\mathcal{L}_p = \mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + pQD_1).$$

On Δ_F , the weighted section products all take values in this same line bundle, because $pD_1 = pD_2$ there. Their value rank at any point of the diagonal $Y' \subset Y' \times Y'$ is at most $h^0(\Delta_F, \mathcal{L}_p)$. The canonical Frobenius filtration [18, Corollary 3.5] has graded pieces

$$\mathcal{G}_j = \mathcal{O}_Y(2B_p + pQD) \otimes \mathcal{A}_j(\Omega_Y^1), \quad 0 \leq j \leq 3(p-1), \quad e_j = \text{rk } \mathcal{A}_j, \quad \sum_j e_j = p^3,$$

Here $\mathcal{A}_j$ is the truncated symmetric power of degree j , locally the degree- j part of

$$\mathcal{O}[u_1, u_2, u_3]/(u_1^p, u_2^p, u_3^p).$$

Write $V = \Omega_Y^1|_C$. Langer's bound [17, Corollary 2.5] gives $L_{\min}(V) \geq -c/(p-1)$ with c depending only on the fixed genus and rank. Put $u = 3(p-1)$. The perfect complementary degree pairing gives

$$\mathcal{A}_{u-j}(V) \simeq \mathcal{A}_j(V)^\vee \otimes (\det V)^{p-1}.$$

Truncated symmetric powers are quotients of tensor powers; their asymptotic least slopes are at least the corresponding multiple of $L_{\min}(V)$. Since $\deg \det V = 0$, the pairing implies $\mu_{\max}(\mathcal{A}_j(V)) \leq 3c$, uniformly in j, p .

Set $\Lambda = l_1 l_2$, so $[C] = \Lambda P^2$ and $h_i C = l_i \Lambda P^3$. The differences

$$B_p - pP = T - (p - k_0 N_p)P$$

belong to a fixed finite list. The displayed intersection hypothesis therefore gives one constant c_0 , independent of j, p , for which

$$\mu_{\max}(\mathcal{G}_j|_C) \leq M_p := 7p\Lambda P^3 + c_0.$$

Successive restriction along the fixed flag yields

$$\begin{aligned} h^0(Y, \mathcal{G}_j) &\leq e_j(M_p + 1) \prod_{i=1}^2 \left(1 + \frac{M_p}{l_i \Lambda P^3}\right) \\ &= e_j p^3 (343P^3 + O(p^{-1})). \end{aligned}$$

Indeed only nonnegative twists (b_1, b_2) with $b_1 h_1 C + b_2 h_2 C \leq M_p$ contribute. On the curve, evaluation at $\lfloor M_p \rfloor + 1$ points bounds each contributing section space by $e_j(M_p + 1)$. All constants are fixed before p varies. Summing the graded pieces gives

$$(12) \quad h^0(\Delta_F, \mathcal{L}_p) \leq R_p(343P^3 + O(p^{-1})) < R_p/4.$$

The last inequality follows from $343P^3 < (14\epsilon)^3 < 1/4$ for large p . There is no extra factor for the number of graded pieces: their total rank is exactly p^3 .

Upper multiplicity bound for the determinant divisor. The Frobenius formula for the first Chern class [19, Lemma 4.2] gives, under the natural numerical identification with the twisted fibre,

$$\frac{c_1(\mathcal{Q}_i)}{R_p} = \frac{B_p}{p} + \frac{p-1}{2p} K_Y + \lambda_i D = P + \lambda_i D + O(p^{-1}), \quad \lambda_i = m_i/R_p \in [0, Q].$$

Twist the specialized blowup and its fixed movable witness as well. For every member $E_i \in |\mathcal{Q}_i|$, its proper transform is effective. Intersecting it with the specialized class γ gives

$$\frac{\text{mult}_{x'}(E_i)}{R_p} \leq 40\epsilon + O(p^{-1}).$$

The fixed complete intersection retains positivity on the specialized model. The intersection inequality for effective divisors therefore applies to every member of these linear systems on the fibre. By Künneth, σ lies in the tensor product of the two section spaces. Choose bases adapted to the filtration by the multiplicity of the corresponding divisors at x' . Their leading forms are independent in each degree, and tensor products remain independent in each bidegree. Consequently

$$(13) \quad \text{mult}_{(x',x')}(E_\sigma) \leq R_p(80\epsilon + O(p^{-1})).$$

Lower multiplicity bound for the same divisor. By (12), the chosen square evaluation matrix has rank less than $R_p/4$ at (x', x') . Constant invertible row operations make more than $3R_p/4$ rows vanish there. Every determinant monomial contains one entry from each of those rows, so

$$(14) \quad \text{mult}_{(x',x')}(E_\sigma) > 3R_p/4.$$

Equations (13) and (14) contradict $80\epsilon < 3/4$ as $p \rightarrow \infty$. $\square$

Completion of the proof of Lemma 1.3. Since $P - r_t D$ is nef,

$$(2P + QD)P^2 \leq (2 + Q/r_t)P^3 < 4P^3.$$

The other numerical hypotheses of Proposition 5.3 hold by our choice of ϵ and t . That proposition excludes precisely the fixed data in the lemma. $\square$

6. THE REMAINING BRANCHES

If $\nu(D) = 0$, then $D \equiv 0$, and Lemma 6.1 gives a nonzero section of a multiple. If $\nu(D) = 3$, then D is big and the conclusion follows. Hence we need only consider

$$D^3 = 0, \quad \nu(D) \in \{1, 2\}.$$

The full-nef branch was the focus of the preceding sections. The smaller-nef branch needs an effective rational numerical representative, followed by the torsion argument below.

Lemma 6.1. *If E is an effective rational divisor on X and $E \equiv D$, then $E \sim_{\mathbb{Q}} D$. Consequently $H^0(X, mD) \neq 0$ for some integer $m > 0$.*

Proof. The connected Picard variety is trivial. Numerically trivial line bundles form $\text{Pic}^\tau(X)$, which has finitely many components modulo $\text{Pic}^0(X)$. They are therefore torsion. Clear the denominators of E and then kill this torsion. Thus $mD \sim mE$ for some integer $m > 0$, and the effective divisor mE supplies a nonzero section. $\square$

Proposition 6.2. *If $n(X, D) < 3$, then D has an effective rational numerical representative, and $H^0(X, mD) \neq 0$ for some $m > 0$.*

Proof. The case of nef dimension zero is already settled. Resolve the almost holomorphic nef reduction, and use birational descent of nef divisors [5, Lemma 3.1]. After modifying the base, it gives

$$X \xleftarrow{p} Y \xrightarrow{f} B, \quad p^*D \equiv f^*M_B,$$

where Y, B are smooth and projective, f has connected fibres, $1 \leq \dim B \leq 2$, and M_B is nef and rational Cartier. Numerical triviality on the general fibre is the hypothesis of this descent lemma; no linear triviality of D on that fibre is being assumed.

Since $K_X \sim 0$ and X is smooth, $K_Y \sim E$ with $E \geq 0$ p -exceptional. A general fibre of f maps birationally to a smooth compact fibre of the nef reduction. Adjunction on its morphism locus makes the canonical bundle of that compact fibre trivial, and E restricts to an exceptional divisor over it. Thus the resolved fibre has canonical Iitaka dimension zero. The general fibre has dimension at most two, so its ordinary canonical divisor has a good minimal model. Relative finite generation is [11, Theorem 1.4]. These are precisely the two hypotheses of [12, Theorem 2.12]; that theorem gives a good model over B , and [12, Corollary 2.9] realizes it by a relative program, as in [3, Sections 8.4–8.5]. Write its relative canonical fibration as

$$Y_2 \xrightarrow{g} B_2 \xrightarrow{j} B, \quad K_{Y_2} \sim_{\mathbb{Q}} g^* A_2, \quad \dim B_2 = \dim B \leq 2.$$

Here Y_2 is klt, g has connected fibres, and A_2 is rational Cartier. The equality of dimensions follows from canonical Iitaka dimension zero on the general fibre over B . Ambro's theorem [6, Theorem 0.2] gives an effective rational boundary Δ such that (B_2, Δ) is klt and

$$A_2 \sim_{\mathbb{Q}} K_{B_2} + \Delta.$$

The base adjoint is pseudo-effective: $\kappa(Y_2, K_{Y_2}) = 0$, and the projection formula for g descends a nonzero canonical section. Apply Lemma 2.1 on B_2 , with the nef divisor $j^* M_B$, to obtain

$$G_{B_2} \geq 0, \quad G_{B_2} \equiv K_{B_2} + \Delta + j^* M_B.$$

Take a common smooth resolution $a : U \rightarrow Y$, $b : U \rightarrow Y_2$. The ordinary relative program and its pullback comparison give

$$a^* K_Y = b^* K_{Y_2} + F, \quad F \geq 0, \quad a^* f^* M_B = b^* g^* j^* M_B.$$

Hence the effective rational divisor

$$G_U = b^* g^* G_{B_2} + F \equiv a^* (K_Y + p^* D) \equiv (pa)^* D + a^* E$$

pushes to $G_X = (pa)_* G_U \geq 0$ with $G_X \equiv D$. Numerical pushforward here is valid because X is smooth; see [5, Lemma 2.14]. The exceptional term $a^* E$ has zero pushforward. Lemma 6.1 finishes the proof. $\square$

Only numerical nonvanishing on the lower-dimensional base was used. The positive-part and numerical-semiampleness constructions in the source are unnecessary for this proposition.

Corollary 6.3. *Let X be a smooth projective threefold over $\mathbb{C}$ with $K_X \sim 0$ and $q(X) = 0$, and let D be a nef Cartier divisor. Then $H^0(X, \mathcal{O}_X(mD)) \neq 0$ for some integer $m > 0$.*

Proof. If $D \equiv 0$, use Lemma 6.1. If $D^3 > 0$, use bigness. The smaller-nef case is Proposition 6.2; the remaining case $\kappa(X, D) = 3$ and $D^3 = 0$ is the opening theorem. Thus this corollary has the same source-dependent proof status as that theorem. $\square$

Disclosure. This AI-assisted report is intended for joint learning, discussion, and checking. Its correctness has not been independently verified. No claim of novelty, priority, or authorship credit is made. Published and preprint inputs are attributed explicitly; the scope of the source-dependent proof draft is stated above.

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